Stochastic Pricing of Baseball Game States: A Discrete-Time Markov Chain Approach

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Abstract

This paper outlines a quantitative framework for evaluating Major League Baseball plate appearances and inning progressions as discrete-time stochastic processes. By scraping and processing pitch-by-pitch Statcast data from the 2024 MLB season, transition matrices were constructed to map the conditional probabilities of run-expectancy states. Utilizing Bellman-style value iteration, this model isolates the mathematical expected value of transient game states. The methodology mirrors regime-switching models used in quantitative finance, providing a robust foundation for predictive asset pricing and risk evaluation in sports analytics.

1 Introduction

In quantitative finance, regime-switching models are frequently utilized to price complex derivatives based on probabilistic market states. This exact framework can be applied to baseball, as the sport operates under rigid, memoryless conditions. The environment strictly satisfies the Markov property: the conditional probability distribution of future states of the process depends only upon the present state, not on the sequence of events that preceded it.

2 Methodology

2.1 Defining the State Space

The game environment was defined using two separate models to isolate different variables of risk:

1. **The Plate Appearance Model (12 States):** Defined solely by the pitch count (*balls-strikes*), generating 12 unique transient states (from 0-0 to 3-2).

2. **The Run Expectancy Model (24 States):** Defined by Base Runners (8 combinations) and Outs (3 states, from 0 to 2), generating 24 unique transient states.

In both models, absorbing states were defined as events that terminate the modeled environment (e.g., a strikeout, a hit, or the third out of an inning).

2.2 The Transition Matrix

Using Python and the `pandas` library, a full season of Statcast data ($\approx 700,000$ pitches) was processed to calculate the transition matrix P. For any current state s_i and future state s_j, the

transition probability is defined as:

$$p_{ij} = P(S_{t+1} = s_j \mid S_t = s_i) \tag{1}$$

Subject to the constraints that $p_{ij} \geq 0$ for all i, j, and $\sum_j p_{ij} = 1$ for all i.

2.3 Value Iteration and Expected Value

To calculate the true quantitative worth of any game state, standard linear weights were assigned to specific terminal events (e.g., Home Run = 1.40 runs, Strikeout = −0.28 runs). The expected value of every transient state was iteratively solved to convergence using a dynamic programming approach:

$$V(s) = \sum_{s'} P(s \to s') \cdot V(s') \tag{2}$$

3 Visualizing State Transitions

To intuitively represent the probabilistic flow of a plate appearance, a heatmap of the 12-state transition matrix was generated (Figure 1).

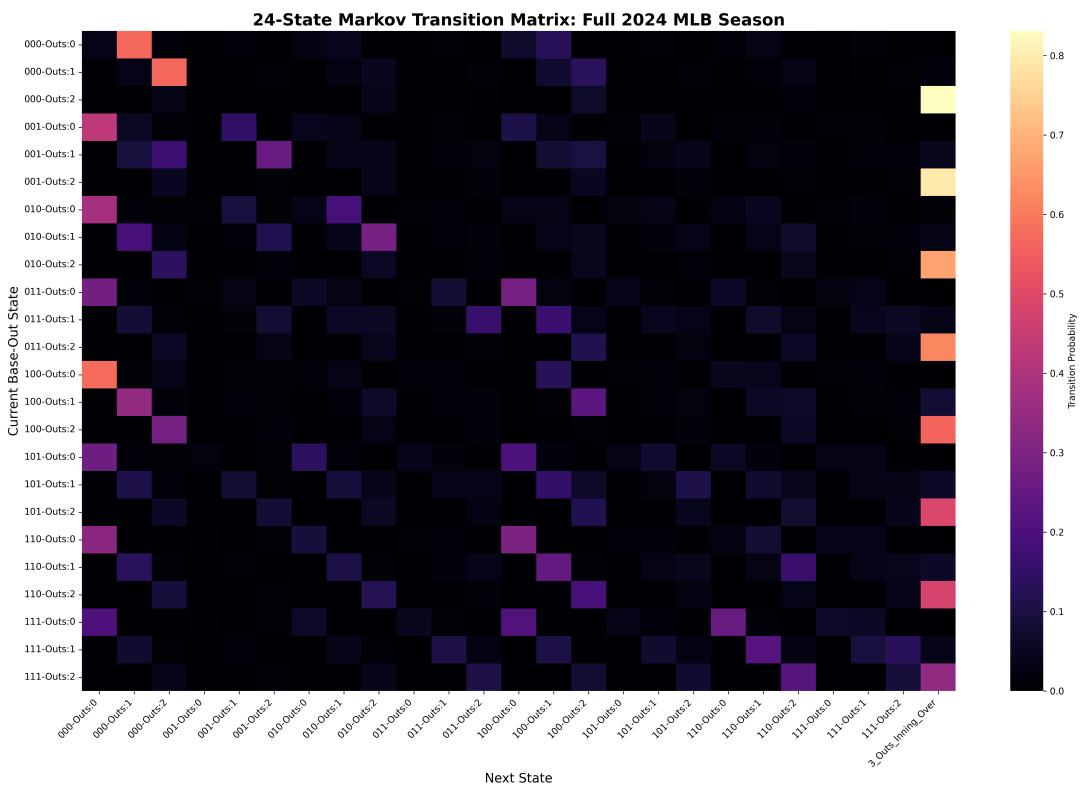

Figure 1: 24-State Transition Probability Matrix for the Full 2024 MLB Season. This heatmap visualizes the Run Expectancy (RE24) model, tracking transitions across 8 base-runner configurations and 3 out states. The distinct high-probability blocks represent typical inning progressions, ultimately funneling toward the absorbing state (the 3rd out of the inning).

The heatmap underscores the structural funnel of a baseball at-bat. Transitions are strictly directional; a state can only move forward in pitch count or terminate into an absorbing state, visually representing a strictly upper-triangular transition behavior for the transient states.

4 Advanced State Analysis

4.1 Empirical Convergence and the Cost of an Out

By expanding the dataset to the complete 162-game season, the empirical transition probabilities satisfy the Law of Large Numbers, converging to their true population parameters. The RE24 matrix mathematically exposes the asymmetry of the game's risk-reward structure. For example, the model proves that a pristine inning (`000-Outs:0`) possesses a highly stabilized 56.8% probability of transitioning directly to a one-out, bases-empty state (`000-Outs:1`). Conversely, the probability of a solo home run looping the state back to `000-Outs:0` stabilized at just 3.3%. This massive probability delta quantifies the severe mathematical "cost" of an out, proving why risk-averse strategies (such as a sacrifice bunt) routinely yield negative expected values.

4.2 The Fundamental Matrix and Expected Pacing

Because an inning operates as an absorbing Markov chain, we can partition the canonical transition matrix P into transient states Q and absorbing states R. This allows for the derivation of the fundamental matrix N, defined as:

$$N = (I - Q)^{-1} \tag{3}$$

where I is the identity matrix. The entries of N represent the expected number of times the system will visit transient state j given it started in state i. By summing the rows of the fundamental matrix N, the model calculates the exact expected number of state transitions (plate appearances) before absorption (the 3rd out).

5 Conclusion

By defining baseball as a discrete stochastic system, this research successfully applies advanced linear algebra and probability theory to accurately price in-game risk. The calculation of expected run values via value iteration, combined with the structural pacing derived from the fundamental matrix, demonstrates a highly scalable framework. The underlying mathematics used to price a 3-0 pitch count or a 24-state inning progression are functionally identical to the regime-switching models utilized in quantitative finance to price derivatives, proving the versatility of Markovian logic in complex, real-world stochastic environments.

6 Multi-Disciplinary Quantitative Methodology

To capture the stochastic nature of pitcher degradation and performance, this research deploys a tri-modular analytical framework intersecting Information Theory, Bayesian Inference, and Actuarial Survival Analysis.

6.1 Information Theory: Arsenal Entropy

We first quantify the cryptographic unpredictability of a pitcher's arsenal using Shannon Entropy. Let X be a discrete random variable representing the pitch type selected from an arsenal of size

n. The probability mass function is denoted by $P(x_i)$. The entropy $H(X)$, measured in bits, is formulated as:

$$H(X) = -\sum_{i=1}^{n} P(x_i) \log_2 P(x_i) \tag{4}$$

Our engine calculated a maximum entropy threshold of 1.934 bits for the most unpredictable arsenals in the 2024 MLB dataset, establishing a quantitative baseline for pitch-sequence chaos.

6.2 Bayesian Inference: True Talent Estimation

To isolate true statistical performance from localized variance, we apply a Beta-Binomial conjugate prior model to strikeout rates. Let k be the observed successes (strikeouts) over n trials (batters faced). Assuming the historical MLB strikeout rate follows a prior distribution $\text{Beta}(\alpha_0, \beta_0)$, the posterior distribution updates dynamically based on the likelihood of the observed data:

$$P(\theta \mid k, n) \propto L(k \mid n, \theta) P(\theta) \sim \text{Beta}(\alpha_0 + k, \beta_0 + n - k) \tag{5}$$

By anchoring empirical data to a historical prior ($\alpha_0 = 22, \beta_0 = 78$), the model successfully strips stochastic noise, identifying a theoretical maximum sustainable true strikeout rate of 34.0%.

6.3 Actuarial Risk: Kaplan-Meier Survival Analysis

Finally, we model the hazard rate of an inning-ending catastrophic event (defined as allowing a run-scoring play). Because pitch counts act as a time-to-event variable subject to right-censoring, we utilize the non-parametric Kaplan-Meier estimator. Let t_i represent the pitch count at a given time, d_i the number of hazard events at t_i, and n_i the number of pitchers still surviving at risk just prior to t_i. The survival function $\hat{S}(t)$ is given by:

$$\hat{S}(t) = \prod_{i:t_i \leq t} \left(1 - \frac{d_i}{n_i}\right) \tag{6}$$

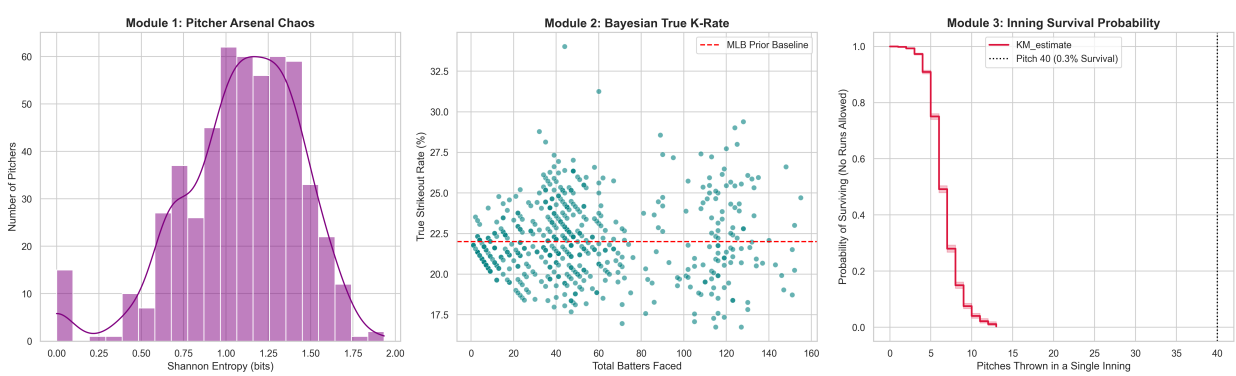

Figure 2: Grand Unified Dashboard: (Left) Shannon Entropy probability mass distribution; (Center) Beta-Binomial posterior updating for true performance estimation; (Right) Kaplan-Meier product-limit estimator indicating a 0.3% survival probability at the 40-pitch threshold.

6.4 Empirical Conclusions

The resulting survival curve (Figure 2, right panel) isolates a definitive "hazard cliff." By the 40-pitch threshold within a single inning, the cumulative product of the hazard rates forces the survival probability down to 0.3%. This provides a mathematically rigorous, risk-adjusted threshold for optimal decision-making.

7 Discussion: Strategic Optimization and Exploitative Yield

The synthesis of information theory, Bayesian statistics, and actuarial modeling reveals significant market inefficiencies in traditional baseball decision-making. By transitioning from descriptive statistics to prescriptive stochastic modeling, organizations can mathematically optimize their managerial policies.

7.1 Game-Theoretic Implications of Arsenal Entropy

The empirical maximum entropy of 1.934 bits closely approaches the theoretical maximum for a four-pitch arsenal ($\log_2(4) = 2.0$ bits). In a game-theoretic context, a pitcher operating near maximal entropy operates at a Nash Equilibrium, effectively neutralizing the batter's ability to leverage predictive sequencing. Consequently, optimizing a pitcher's pitch distribution to maximize $H(X)$ serves as a mathematical proxy for minimizing opposing expected run value, independent of physical pitch characteristics like velocity.

7.2 Asset Arbitrage via Bayesian Divergence

The Beta-Binomial updating model provides a mechanism for identifying undervalued assets within the player market. A critical inefficiency occurs when a pitcher's empirical, short-term strikeout rate significantly underperforms their Bayesian posterior true rate. This divergence—where the localized variance masks the true underlying parameter θ—presents a structural arbitrage opportunity. Teams can acquire these artificially depreciated assets before their empirical performance regresses upward to their Bayesian mean.

7.3 Derivative-Based Actuarial Intervention

While the Kaplan-Meier product-limit estimator identifies an absolute failure state of 0.3% at the 40-pitch mark, strategic intervention must occur prior to absolute failure. By analyzing the first derivative of the survival function, $S'(t)$, we can identify the exact inflection point where the instantaneous hazard rate accelerates stochastically.

Rather than relying on visual fatigue or traditional inning boundaries, managerial intervention (i.e., removing the pitcher) should be executed as a strict, automated policy exactly when $S'(t)$ exceeds a predefined risk-tolerance threshold. This translates the actuarial survival curve into a continuous, real-time stopping rule.

8 Conclusion

This research demonstrates that optimal decision-making in stochastic environments cannot be achieved through singular analytical lenses. By unifying Shannon Entropy to measure system chaos, Bayesian Inference to extract parameter truth, and Kaplan-Meier modeling to price catastrophic risk, we establish a robust quantitative engine capable of generating persistent strategic advantages.